Bonds and stocks CHEAT SHEET

WARM-UP: PRESENT VALUE

$PV = \frac{C}{(1+r)^t}$: $1,000 in 4 years at 5 % → $822.70

perpetuity $\frac{C}{r}$: $60 a year forever at 5 % → $1,200

annuity factor $\frac{1}{r} - \frac{1}{r(1+r)^t}$: 4 years at 5 % → 3.546

$60 × 3.546 = $212.76 (edges: r → 0 gives t; t → ∞ gives 1/r)

A BOND'S PRICE

a bond: face value · coupon · maturity → a fixed list of payments

$$\text{price} = \text{coupon} \times \text{annuity factor} + \frac{\text{face}}{(1+r)^t}$$

an annuity plus one final payment

4 years, 6 % coupon, at 5 %: 212.76 + 822.70 = 1,035.46

coupon rate > yield: premium; < yield: discount (4 % → 964.54)

edges: coupon rate = yield → face value; r = 0 → 1,240 (the sum)

two coupons a year: 8 periods of $30 at 2.5 % → 1,035.85

YIELD TO MATURITY

yield to maturity: the one rate that makes

PV(payments) = price. Found by trial, or RATE / YIELD

price 966.13: 6 % → 1,000 · 8 % → 933.76 · 7 % → 966.13

current yield = coupon / price = 6.21 % (income only)

earned only if held to maturity and paid in full

DURATION: HOW PRICES MOVE

when yields rise, prices fall: 5 % 1,035.46 · 6 % 1,000 · 7 % 966.13

6 % → 7 %: 4-year bond −3.39 %; 30-year bond −12.41 %

$$\text{duration} = \Sigma\ \text{year} \times \frac{\text{PV(payment)}}{\text{price}}$$ (the average wait)

shares 5.52 · 5.26 · 5.01 · 84.22 % → 3.68 years

edges: zero-coupon → maturity (4); 12 % coupon → 3.47

$$\text{modified duration} = \frac{\text{duration}}{1+y} = 3.50$$

% change in price ≈ − modified duration × change in yield

5 % → 6 %: rule −3.50 %, exact −3.42 % (the curve bends)

SPOT RATES AND THE LAW OF ONE PRICE

spot rate: the rate for one payment at one date

$PV = \Sigma\ \frac{C_t}{(1+r_t)^t}$ each payment at its own rate

3 · 3.5 · 4 · 4.5 %: 58.25 + 56.01 + 53.34 + 888.88 = 1,056.48

its yield 4.43 %: a blend, close to the 4-year rate

law of one price: same cash, same date → same price

bond at 1,040 → buy it, sell the pieces: +16.48, no risk

term structure: expected rates + a reward for waiting

REAL RATES AND DEFAULT

$(1 + \text{real}) = \frac{1+\text{nominal}}{1+\text{inflation}}$: 5 % and 2 % → 2.94 %

shortcut 5 − 2 = 3 %: a little high

indexed bond: coupons and face rise with prices

20 · 20 · 1,020 real → 20.80 · 21.63 · 1,147.36 at 4 %

real at 2 % = nominal at 6.08 % = 1,000

default: promise 1,050; 4 % chance of 500 → expect 1,028

price 988.46 · promised yield 6.23 % · spread 2.23

the promised yield is the best case, not the average

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RETURNS AND THE DIVIDEND DISCOUNT MODEL

expected return $= \frac{DIV_1 + P_1 - P_0}{P_0} = (3 + 52 - 50)/50 = 10\,\%$

$=$ dividend yield 6 % + capital gain 4 %

$P_0 = \frac{DIV_1 + P_1}{1 + r} = 55/1.10 = 50$ (r: the cost of equity)

dividend discount model: P_0 = PV of all future dividends

3-year horizon: 7.74 + 42.26 = 50

the far price fades: year 10 → 28.53 · 50 → 3.03 · 100 → 0.18

CONSTANT GROWTH AND THE COST OF EQUITY

constant growth $P_0 = \frac{DIV_1}{r - g}$: $3/(0.10 - 0.04) = 50$

(keep g in the fund, pay out r − g)

edges: g = 0 → 30; g = 9 % → 300; g → r: no limit (needs g < r)

cost of equity $r = \frac{DIV_1}{P_0} + g = 6\,\% + 4\,\% = 10\,\%$

g = plowback × return on equity = 0.4 × 10 % = 4 %

EPS 5 → dividend 3, kept 2; stake 50 → 52; EPS → 5.20

GROWTH OPPORTUNITIES AND FREE CASH FLOW

$P_0 = \frac{EPS_1}{r} + PVGO$

new investment earns 10 % = r: 5/0.10 = 50, PVGO = 0

earns 15 %: g = 6 %, price 75, PVGO = 25 (P/E 15, not 10)

earns 8 %: g = 3.2 %, price 44.12, PVGO = −5.88

growth adds value only when new investment earns more than r

free cash flow $-0.8 \cdot 0.6 \cdot 1.6 \cdot 2.4 \cdot 3.0$, then +4 %, at 10 %

horizon value $= \frac{FCF_6}{r - g} = 3.12/0.06 = 52$

value 4.47 + 32.29 = 36.76: 88 % sits beyond the horizon

CASE: BANGALORE COLD CHAIN (FICTIONAL)

$80M ten-year bond; lenders want 5 % (government 4 %)

4.5 % coupon → 961.39 a bond: raises 76.91, 3.09 short

5 % coupon → 386.09 + 613.91 = 1,000; spread 1 point

real yield 1.94 %; duration 8.11, modified 7.72

at 6 %: 926.40 (−7.36 %): the lenders bear it

share: 0.50/(0.10 − 0.05) = 10.00; g = 37.5 % × 13⅓ % = 5 %

0.80/0.10 = 8.00 + PVGO 2.00; equity 320 + bonds 80 = 400

SPREADSHEET AND PYTHON

PV(rate, n, coupon, face, 0) → the price, with a minus sign

RATE(n, coupon, −price, face, 0) → the yield to maturity

PRICE(settlement, maturity, coupon rate, yield, 100, freq, 0)

YIELD(settlement, maturity, coupon rate, price, 100, freq, 0)

DURATION(settlement, maturity, coupon rate, yield, freq, 0)

MDURATION(…same…) = DURATION / (1 + yield/freq)

dates with DATE(y, m, d); PRICE and YIELD work per 100 of face

freq: coupons a year (1, 2 or 4); the last 0: 30/360 day count

Python: sum(c / (1 + y) ** t for t, c in enumerate(flows, 1))

CHECK BEFORE YOU TRUST IT

before you trust a price:

1 what cash will it really pay, and when? (draw the timeline)

2 what could investors earn elsewhere, at the same risk?

3 is the yield a promise, or what you expect on average?

4 how much of the price is growth that has not happened yet?